

Lecture 2: FEMUR

Proximal End of Femur

Head of Femur

The head of the femur articulates with the acetabulum to form the hip joint. On its surface is a small pit called the **fovea capitis**. This fovea gives attachment to the **ligament of the head of femur (ligamentum teres)**. The ligament attaches to the **non-articular part of the acetabulum** (acetabular notch and transverse acetabular ligament). Although this ligament contributes little to stability, it carries a small artery to the head of the femur, which is more important in children.



Trochanters

The femur has two prominent projections called **trochanters**, a feature unique to the femur.

- **Greater trochanter** lies on the **anterolateral** aspect and serves as an important site for attachment of several powerful muscles (gluteus medius, gluteus minimus, piriformis, obturator internus, etc.).
- **Lesser trochanter** lies on the **posteromedial** aspect and gives attachment mainly to the **iliopsoas muscle**, the chief flexor of the hip.

Trochanters are specialized for **muscle attachment**, helping in leverage and movement at the hip joint.

Intertrochanteric Region

❖ Anterior Aspect – Intertrochanteric Line

On the anterior surface, the **intertrochanteric line** runs between the greater and lesser trochanters. It provides attachment to the **capsule of the hip joint**. Because the capsule extends up to this line, structures anterior to it are considered **intracapsular**. The strong **iliofemoral ligament** (the strongest ligament of the body) is attached along this line and prevents hyperextension of the hip joint.

❖ Posterior Aspect – Intertrochanteric Crest

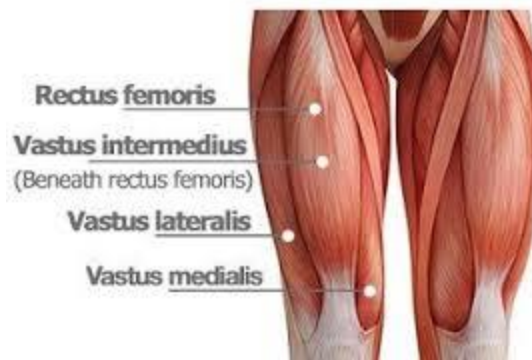
On the posterior surface, the corresponding ridge is the **intertrochanteric crest**.

- ✓ Unlike the intertrochanteric line, it **does not give attachment to the capsule of the hip joint**. The hip joint capsule lies **medial to the crest**, which explains why the **entire neck of the femur is not intracapsular posteriorly**. This distinction is clinically important in fractures of the femoral neck.

Shaft of Femur – Anterior Aspect

The **anterior surface of the femoral shaft is smooth**. It mainly provides attachment to the fleshy parts of the **quadriceps femoris muscle group**.

- **Vastus medialis** arises from the medial part of the anterior femur.
- **Vastus lateralis** arises from the lateral part.
- **Vastus intermedius** arises from the anterior and lateral surfaces of the shaft, lying deep to rectus femoris.



The word **vastus** means **broad**, reflecting the wide muscular origin.

Over the vastus intermedius lies the **rectus femoris**, which originates from the **anterior inferior iliac spine (AIIS)** and crosses both the hip and knee joints, acting as a hip flexor and knee extensor.

Hilton's Law

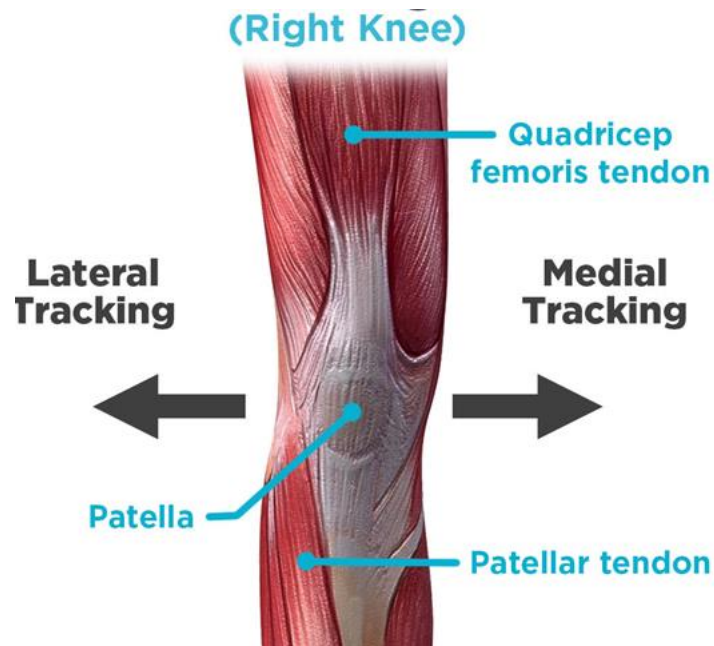
Hilton's law states that **the nerve supplying a joint also supplies the muscles that move the joint and the skin overlying the joint**. This explains why pain from a joint can be felt in the surrounding muscles or skin.

Distal End of Femur – Anterior Aspect

The distal femur presents the **patellar surface**, which articulates with the patella to form the **patellofemoral joint**. This joint is a **plane (gliding) synovial joint**, allowing smooth movement of the patella during knee flexion and extension.

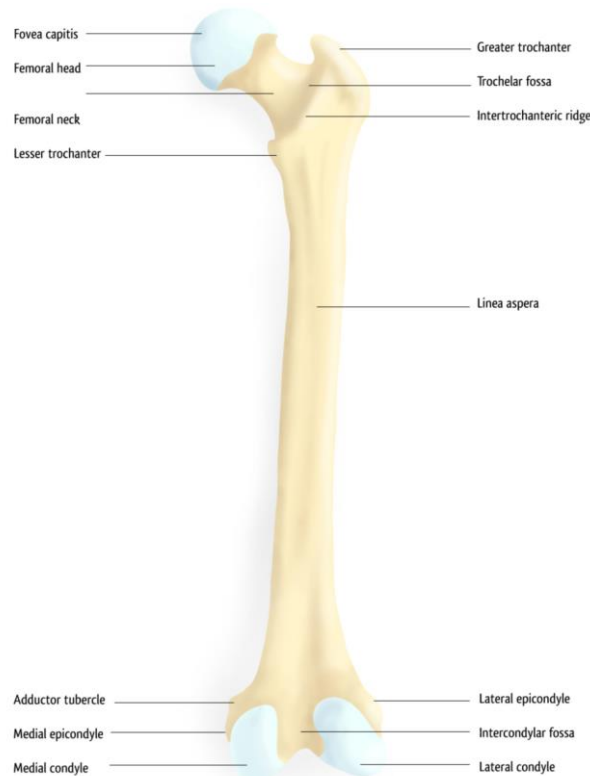
Limits of Patellar Movement

- Superior movement is limited by the **quadriceps femoris tendon**.
- Inferior movement is limited by the **patellar ligament**.
- Medial movement is limited by the tendon of **vastus medialis**.
- Lateral movement is limited by the tendon of **vastus lateralis**.



During the **sitting position (knee flexion)**, the quadriceps muscles relax, allowing the **patella to move superiorly**.

Posterior Aspect of Femur



F. Gaillard
2009
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The posterior surface of the femur is **rough**, reflecting extensive muscle attachment.

Linea Aspera

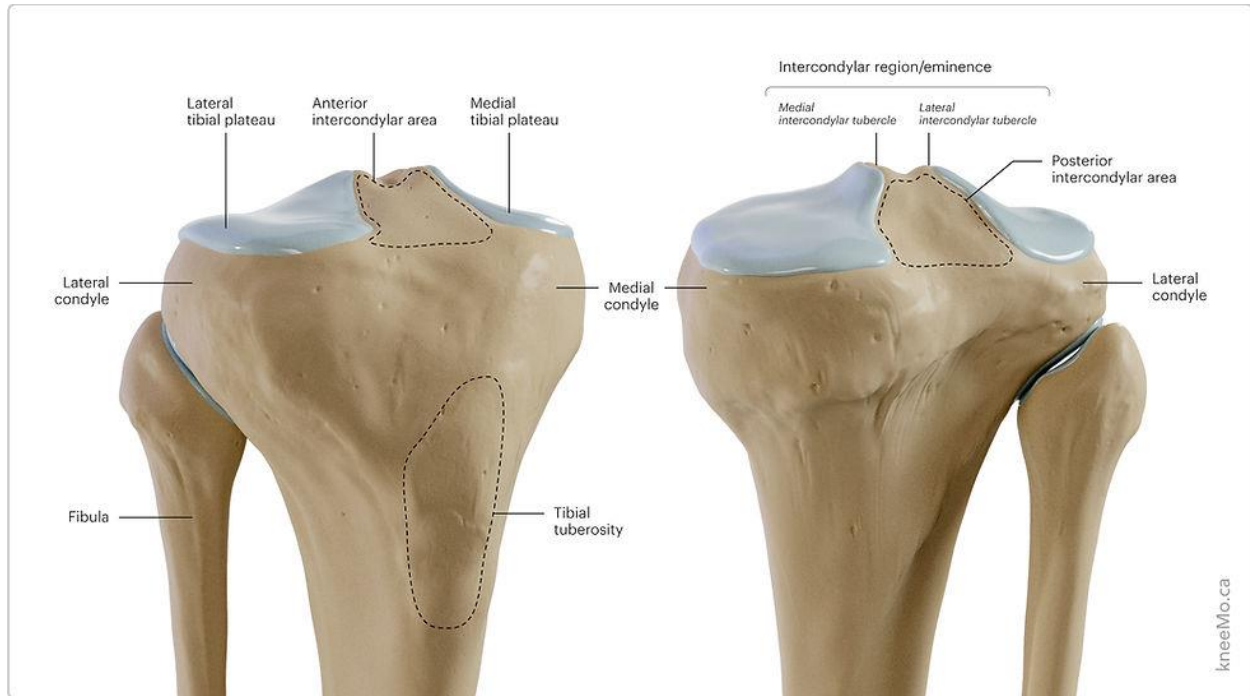
The **linea aspera** is a prominent longitudinal ridge on the posterior shaft. It is extremely important as it provides attachment to many muscles, especially the **adductor magnus**, along with other adductors and short head of biceps femoris.

Intertrochanteric Crest and Fossa

Near the upper posterior femur, the **intertrochanteric crest** extends from the greater trochanter toward the lesser trochanter. Adjacent to it is the **intertrochanteric fossa**, which gives attachment to **only one muscle** – **obturator externus**. There is no comparable muscular attachment directly on the crest itself.

Intercondylar Region

The **intercondylar fossa** lies between the medial and lateral condyles. It receives the **intercondylar eminence of the tibia** and provides space for cruciate ligaments. It is analogous to the olecranon fossa of the humerus in concept, but **functionally different, as it allows articulation rather than deep locking**

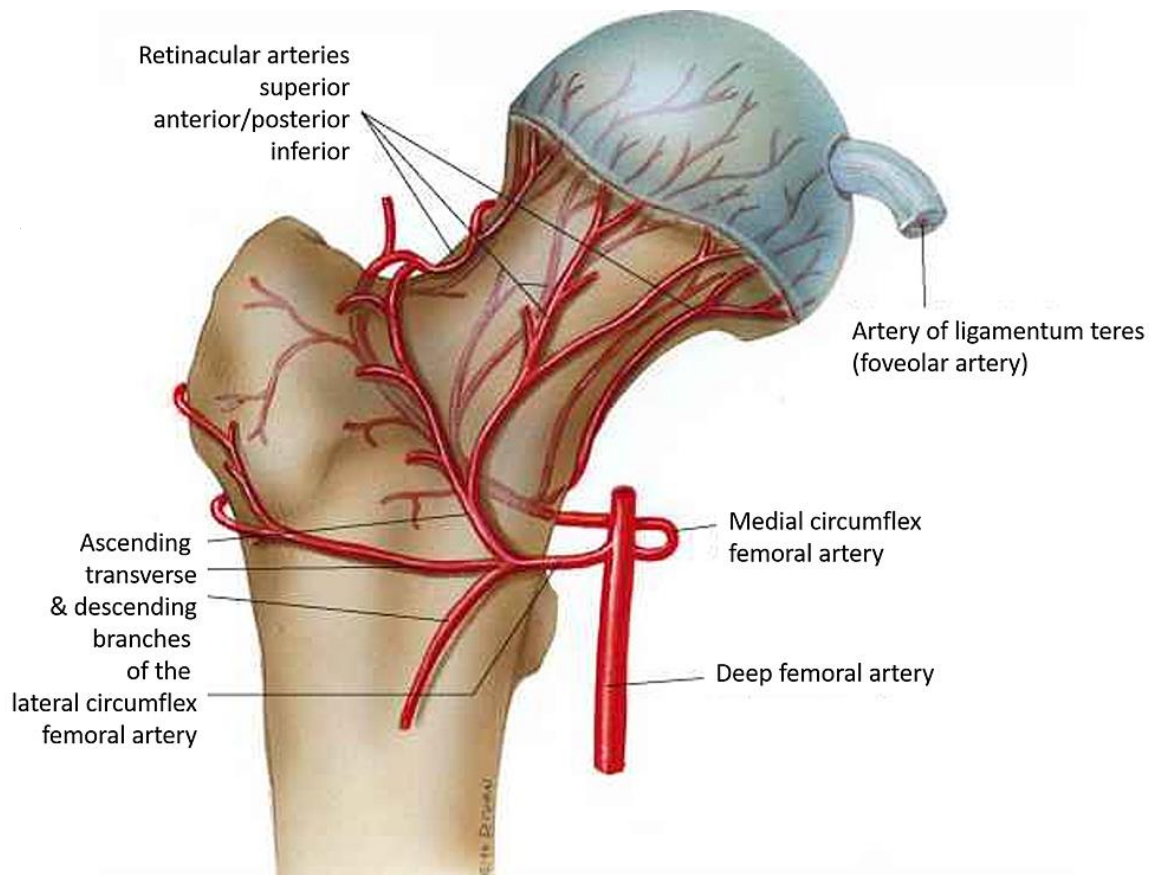


Neck of Femur – Clinical Importance

The **neck of the femur is the weakest part** of the bone and is prone to fractures, especially in the elderly.

The **deep femoral artery** gives rise to the **medial and lateral circumflex femoral arteries**.

- The **medial circumflex femoral artery** is the main blood supply to the head and neck of the femur.
- Its branches, known as **retinacular arteries**, run along the neck of the femur beneath the capsule.



In a **transverse fracture of the femoral neck**, these retinacular arteries may be torn, leading to **avascular necrosis of the femoral head** due to loss of blood supply. This is a major orthopedic emergency and explains the poor healing of intracapsular femoral neck fractures.

Avascular Necrosis follows

- Fracture neck of femur → retinacular artery damage
- Necrosis of head of femur follows
- Artery from obturator artery (via ligament of head)
- Blood supply **insufficient** to prevent necrosis

Angle of Inclination & Angle of Torsion (Femur)

1. Angle of Inclination

This is already familiar, but let's lock it in properly.

The **angle of inclination** is the angle between

- **the long axis of the femoral neck and head, and**
- **the long axis of the femoral shaft.**

In adults, it is normally about 125°.

Functional importance:

This angle allows the femur to place the shaft away from the pelvis, improving balance and efficiency during walking.

2. Angle of Torsion (Femoral Torsion / Version)

The **angle of torsion** describes the **twist** in the femur.

It is the angle between:

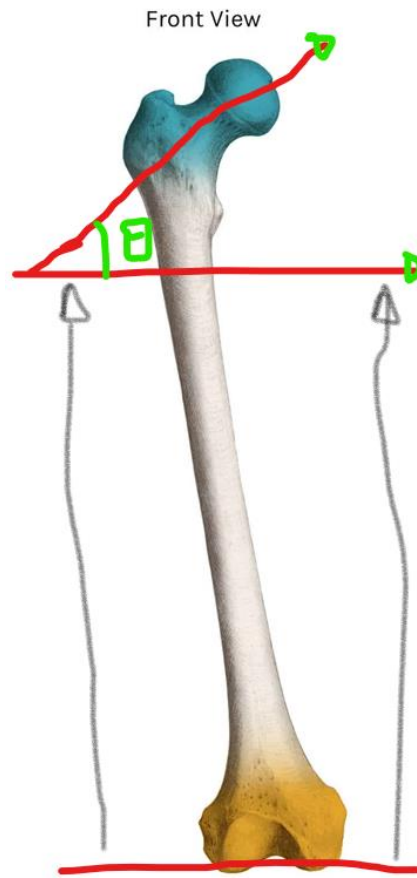
1. A line passing through the **head and neck of the femur**, and
2. A **horizontal line joining the medial and lateral femoral condyles.**

The angle is measured where these two imaginary lines intersect.

Normal value:

- **10°–15° (anteversion)** in adults

*This means the femoral neck is directed slightly **anteriorly** relative to the condyles.*



Relation to Sex and Pelvis

- The angle of torsion is generally **higher in females**.

Reason: females have a **wider pelvis**, which increases the **obliquity of the femoral shaft**.

Key relationship to remember:

- Angle of torsion (θ) is **inversely proportional** to femoral shaft obliquity
- Femoral shaft obliquity is **directly proportional** to pelvic width

So:

Wider pelvis $\rightarrow$ more obliquity $\rightarrow$ higher torsion angle

Effect of Age

- **As age increases** → **angle of torsion decreases**
- *This happens due to **bone remodeling and weight-bearing forces** during growth.*

Functional Significance of Angle of Torsion

- *When θ is small, **abduction of the hip is limited***
- *Proper torsion allows efficient hip movements and stable gait*

Abnormal Angles of Inclination

Coxa Vara ("coxa" means "hip"; Vara (from varus) means "bent inward" or "bow-legged")

- **Excessively decreased angle of inclination**
- *Femoral neck becomes more horizontal*

Seen in:

- **Bow legs (genu varum)** → *excess medial angulation*
- *Leads to increased stress on the femoral neck*

Coxa Valga ("coxa" means "hip"; Valgum (from valgus) means "bent outward" or "knock-kneed.")

- **Excessively increased angle of inclination**
- *Femoral neck becomes more vertical*

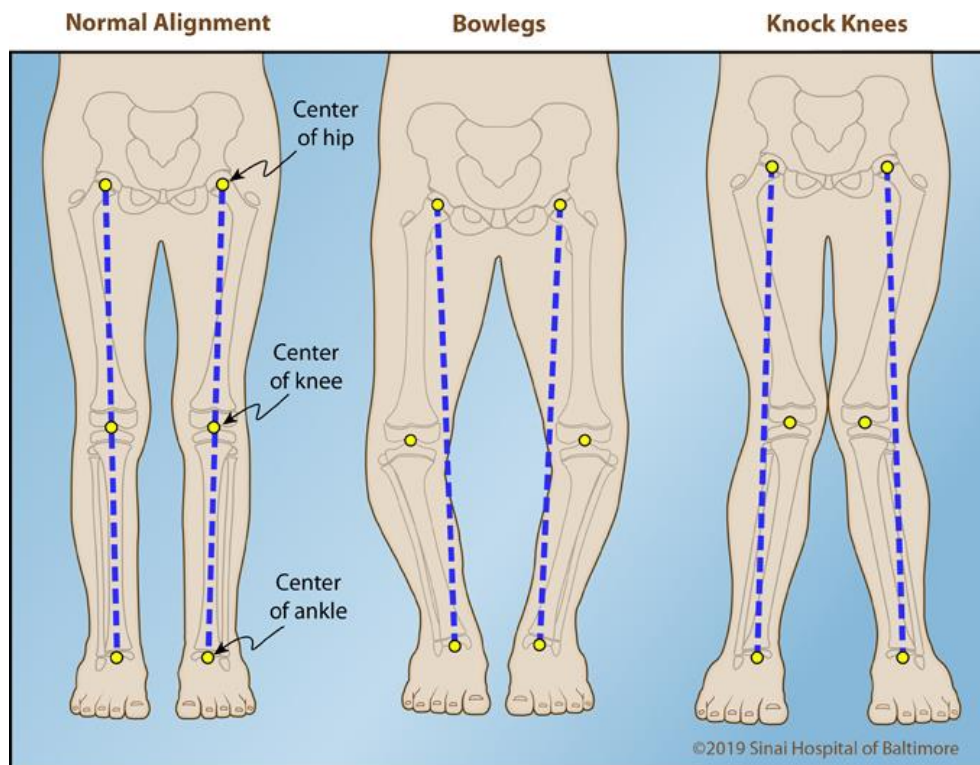
Seen in:

- **Knock knee (genu valgum)**
- **Polio**

Coxa Valga in Polio (Important Explanation)

- Polio damages **lower motor neurons**
- This causes **muscle weakness and paresis** around the hip
- During childhood growth:
 - Muscles fail to exert normal forces on bone
 - Growth becomes **unopposed and abnormal**
 - Femoral neck remains **more vertical**
 - Result → **Coxa Valga**

This is a classic **developmental deformity due to altered muscle pull.**



First, focus on the **blue dashed line** in each limb.
That line is the **mechanical axis** of the lower limb: it joins **center of hip → center of knee → center of ankle**.
Where that line passes tells you how forces travel through the bones.

1. Normal alignment (left image)

The dashed line goes straight:

- Hip → knee → ankle all line up.
- Weight passes evenly through the femur.
- The femoral neck sits at a normal angle (~125°).

Nothing dramatic happens. Bone remodeling stays neutral.

2. Bow legs / Genu varum (middle image)

Now look at the bow-leg picture.

What you should notice:

- The **knees are far apart**.
- The dashed line passes **medial to the knee**.
- The femoral shaft angles **toward the midline** as it goes upward. (Just go up from distal to proximal, what happens to the shaft? It will go to more medial, right?)

This medial shift means:

- More compressive force acts on the **inferior aspect of the femoral neck**.
- During growth, bone responds to this force by making the neck **more horizontal**.

More horizontal neck = **decreased angle of inclination**

Decreased angle of inclination = **coxa vara**

So from the picture:

Bow legs → medial mechanical axis → horizontal neck → **coxa vara**

3. Knock knees / Genu valgum (right image)

Now compare with knock knees.

Here:

- Knees are close together.
- Dashed line passes **lateral to the knee**.
- Femoral shaft is forced **laterally**.

This causes:

- Reduced medial loading
- Femoral neck remodels to become **more vertical**

More vertical neck = **increased angle of inclination**

Increased angle of inclination = **coxa valga**

So:

*Knock knees → lateral mechanical axis → vertical neck → **coxa valga***

“Don’t confuse knocked knee with Locked knee! A **locked knee** is a state in which the knee becomes **stable and fixed in full extension**, requiring little or no muscular effort to maintain the standing position. **As appeared during quiz exam.**”

Fracture of Femur (Distal Part)—Clinically

When the femur fractures distally:

- The **proximal fragment** is pulled **upward** by strong skeletal muscle contraction
- The **distal fragment** is displaced **posteriorly (and upward)**

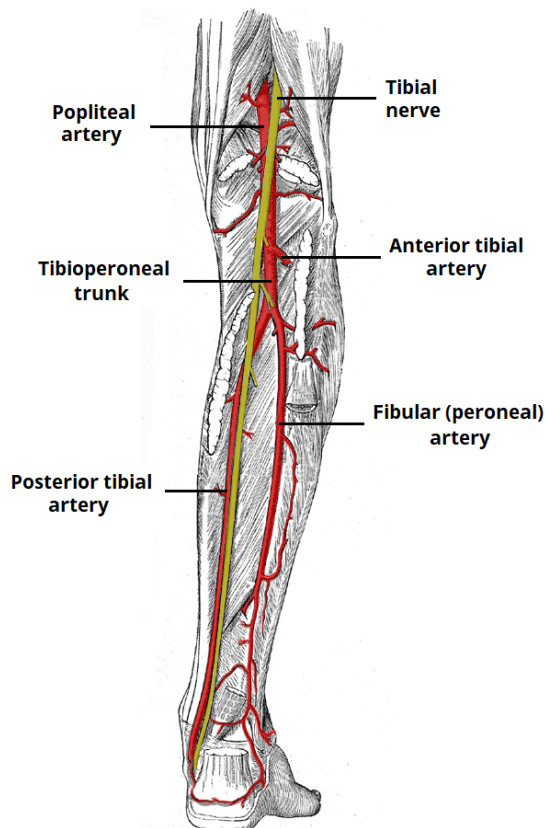
This displacement pattern is important clinically due to:

- Risk to structures in the **popliteal region**

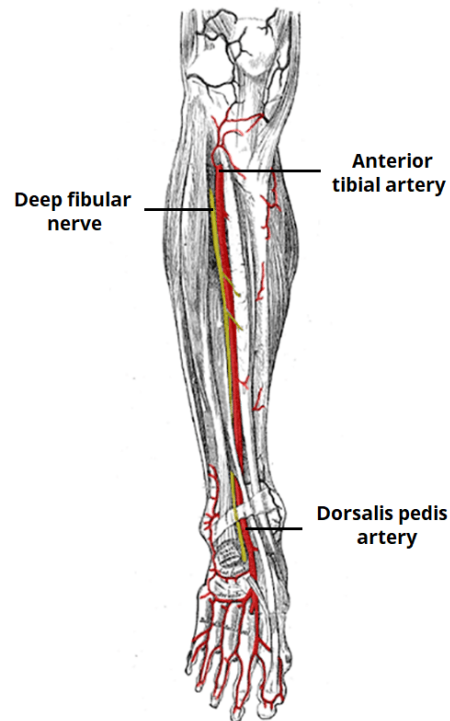
Popliteal Artery (Linked Concept)

- *The popliteal artery:*
 - Supplies the **knee joint**
 - Contributes to **genicular anastomosis**
 - Divides into **anterior tibial** and **posterior tibial arteries**
 - Fibular artery also branches from this artery
 - *It indirectly supplies:*
 - Hamstrings
 - Adductor magnus
 - Calf muscles

This all will be cooked! (In distal fracture of femur)



(i) Posterior Leg



(ii) Anterior Leg

Patella

The **patella** is the **largest sesamoid bone** in the body. It is embedded within the **quadriceps femoris tendon**, which improves the efficiency of knee extension by increasing the leverage of the quadriceps muscle.

Surfaces

The patella has **two surfaces**:

Anterior (outer) surface

- *Convex*
- *Rough*
- *Non-articular*
- *Gives attachment to tendinous expansions of quadriceps*

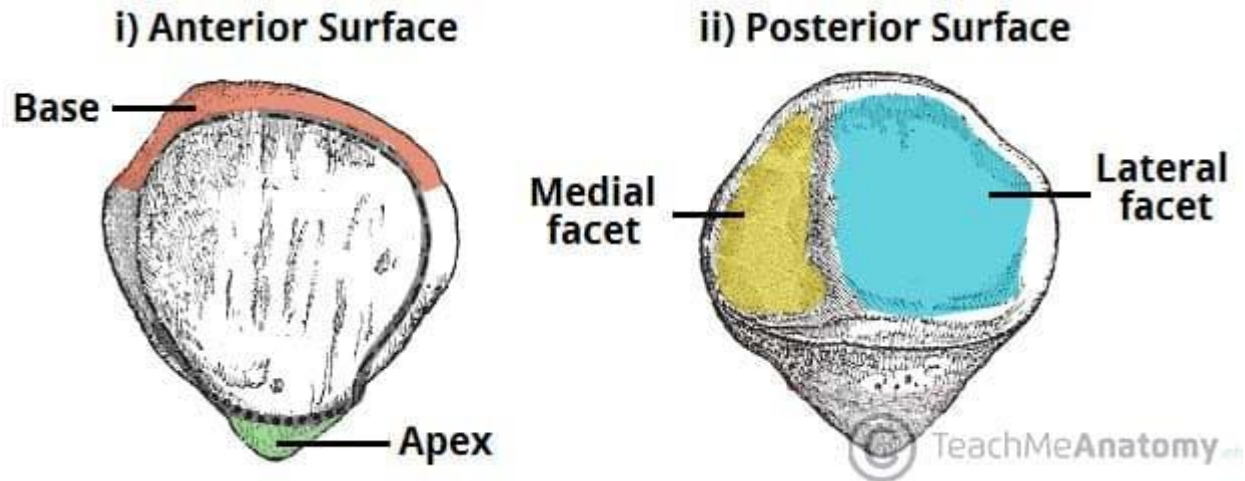
Posterior (inner) surface

- *Concave*
- *Smooth*
- *Articular*
- *Articulates with the femoral condyles*

Borders and Angles

- **Base** – superior border
- **Apex** – inferior pointed end

The **medial and lateral borders** converge inferiorly to form the **apex**. (The lateral articular facet is larger, which helps prevent lateral dislocation.)



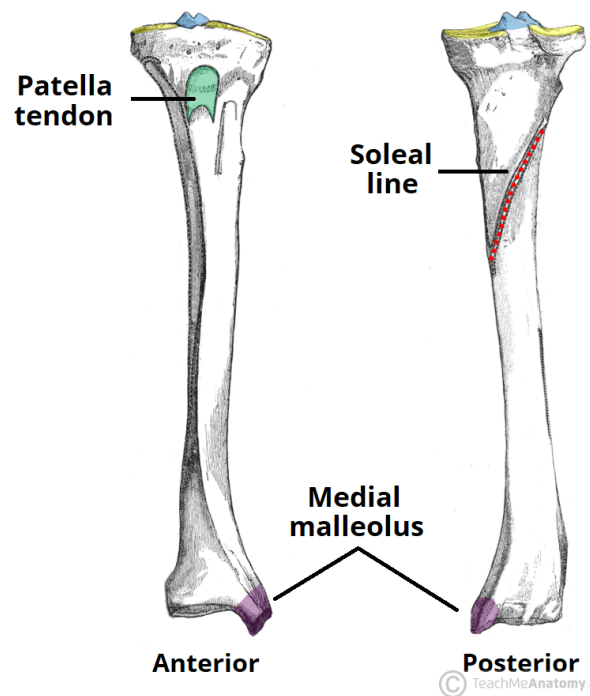
Tendon Relationships

- *Superiorly: **Quadriceps femoris tendon** attaches to the base*
- *Inferiorly: tendon continues as **patellar tendon (ligament)***
- *Patellar tendon inserts into the **tibial tuberosity***

Clinical importance

- *Tapping the **patellar tendon** elicits the **knee jerk reflex***
- *This tests the integrity of:*
 - **Femoral nerve**
 - **L2–L4 spinal segments**

Tibia



The **tibia** is a **large, medial, weight-bearing bone** of the leg.

Shaft features

- **Anterior border** – sharp, subcutaneous → called the **anterior crest**
- **Anterior tibial tuberosity** – palpable; attachment of patellar tendon
- **Medial surface** – subcutaneous and palpable
- **Posterior surface** – smooth, shows the **soleal line** (origin of soleus)

Borders of the Tibia

The tibial shaft has **three borders**:

1. **Anterior border (crest)** – the sharp edge you can feel under the skin; also called the tibial crest.
2. **Medial border** – runs along the medial side, subcutaneous.
3. **Interosseous border** – **lateral border**, where the tibia articulates with the fibula via the interosseous membrane.

Surfaces of the Tibia

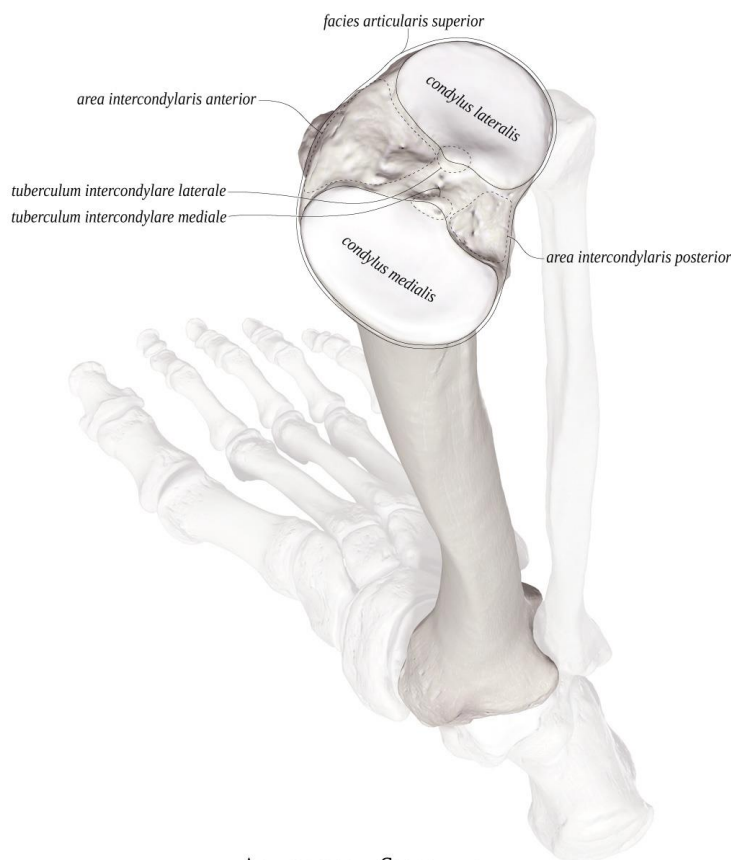
The tibial shaft has **three surfaces**, corresponding to the spaces between the borders:

1. **Medial surface** – smooth, subcutaneous, broad.
2. **Lateral surface** – located between anterior and interosseous borders; slightly convex.
3. **Posterior surface** – between medial and interosseous borders; gives origin to some deep posterior muscles.

There is **no anterior surface**, only an **anterior border**.

Upper end

- **Medial and lateral condyles** – articulate with femoral condyles
- **Intercondylar eminence** with medial & lateral tubercles



Lower end

- **Medial malleolus** – downward projection from medial side
- Articulates with the **talus** at the ankle joint

Articulations

- Proximally – femur
- Distally – talus
- Laterally – fibula (proximal and distal tibiofibular joints)

Fracture point

- Shaft is **narrowest at the junction of middle and lower third**
- This is the **most common fracture site**

Fibula

The **fibula** is a **long, slender, non-weight-bearing bone**.
Its main function is **muscle attachment and ankle stability**.

Borders of the Fibula

The fibular shaft has **three borders**:

1. **Anterior border** – sharp, runs down the front of the shaft.
2. **Posterior border** – less prominent, on the back of the shaft.
3. **Interosseous border** – **medial border**, where it connects to the tibia via the interosseous membrane.

Surfaces of the Fibula

The shaft has **three surfaces**, corresponding to the spaces between the borders:

1. **Lateral surface** – broadest, subcutaneous.
2. **Medial surface** – faces the tibia; gives origin to some muscles.
3. **Posterior surface** – between the posterior and interosseous borders; attachment site for deep posterior muscles.

Proximal end

- **Head and neck of fibula**
 - **Does not articulate with the femur**
- ☐ The **neck of fibula** is a common fracture site.

Distal end

- **Forms the lateral malleolus**
- **Fractures commonly occur just proximal to the lateral malleolus**
- **Often associated with ankle joint injuries**

Movements of the Foot

- **Ankle (talocrural) joint**
 - **Hinge joint**
 - Allows **dorsiflexion and plantarflexion only**
- **Subtalar joint (talocalcaneal)**
 - Allows **inversion and eversion**
- **Transverse tarsal joint**
 - Combination of:
 - Talonavicular joint
 - Calcaneocuboid joint
 - Assists inversion and eversion

Interosseous Membrane

- Located between **tibia and fibula**
- Does **not completely cover** the space between them
- Has:
 - **Proximal opening** → for **deep fibular nerve**
 - **Distal opening** → for **perforating branch of peroneal artery**

Sciatic Nerve and Its Divisions

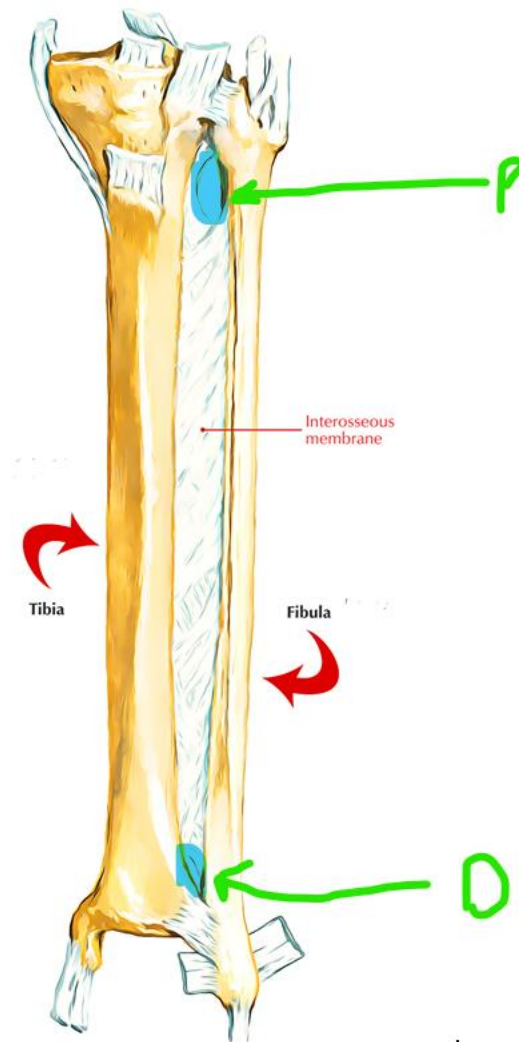
The **sciatic nerve** divides into:

1. **Tibial nerve**
2. **Common fibular (peroneal) nerve**

Both enter the **popliteal fossa**.

Common Fibular Nerve (Branch of sciatic nv)

- Winds around the **neck of fibula**
 - Divides into:
 - **Deep fibular nerve** → anterior compartment
 - **Superficial fibular nerve** → lateral compartment
- Deep fibular nerve passes from **posterior** (Since sciatic nerve is **posterior**) to anterior through the **proximal defect** in the interosseous membrane.



Injury at Neck of Fibula

Fracture at the **head/neck of fibula** injures the **common fibular nerve**.

Paralyzed muscles

- **Dorsiflexors of foot** (anterior compartment)
- **Evertors of foot** (lateral compartment)

Muscles intact

- **Posterior compartment muscles** → *plantarflexion preserved*

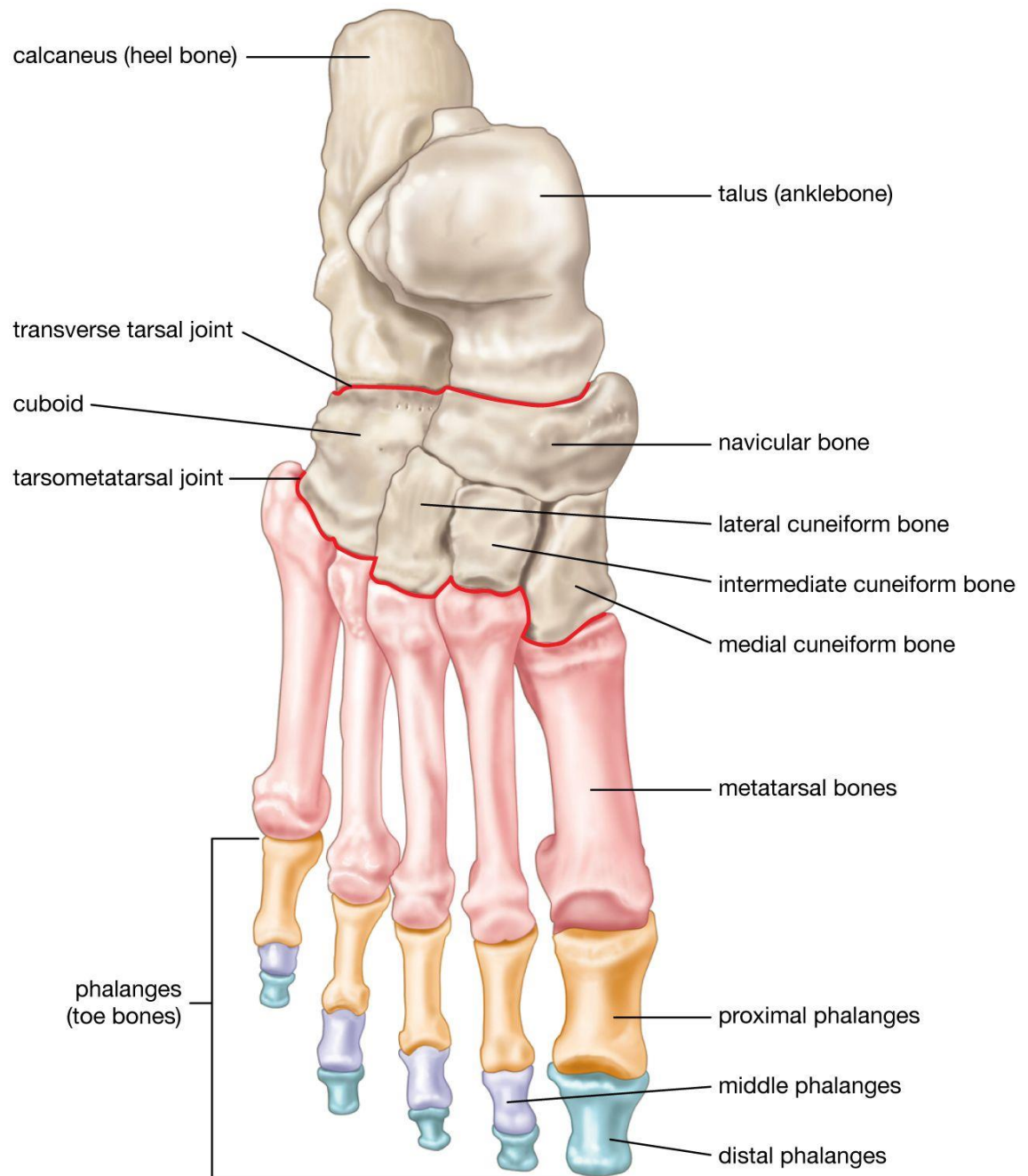
Result

- **Foot drop**
 - Loss of dorsiflexion
 - High-stepping gait



Bones of the Foot

There are **26 bones** in the foot (27 in hand)



Tarsal bones (7)

The foot skeleton is divided into **three parts**:

1. **Hindfoot**

- *Talus*
- *Calcaneus*

2. **Midfoot**

- *Navicular*
- *Cuboid*
- *Three cuneiforms (medial, intermediate, lateral)*

3. **Forefoot**

- *Metatarsals*
- *Phalanges*

Foot Joints & Ligaments

1. **Joints between Hindfoot and Midfoot**

- **Calcaneocuboidal joint**
- **Talonavicular joint**
→ Together, these form the **Transverse Tarsal Joint**, which allows inversion and eversion of the foot.

2. **Ligaments stabilizing the Talocrural (Ankle) Joint**

- **Lateral Collateral Ligament**
 - Includes anterior talofibular, posterior talofibular, and calcaneofibular ligaments.
- **Medial Collateral Ligament (Deltoid Ligament)**
- **General Collateral Ligaments** (support sides of the joint)

Mechanism of Injury:

- Force applied from the **medial side of the ankle**:
 - Tension occurs on the **lateral collateral ligament**, which can be damaged.
 - Talus is pushed **laterally**.

- On anatomical examination, **increased joint space (clear space)** may be visible.

Fractures of the Femur

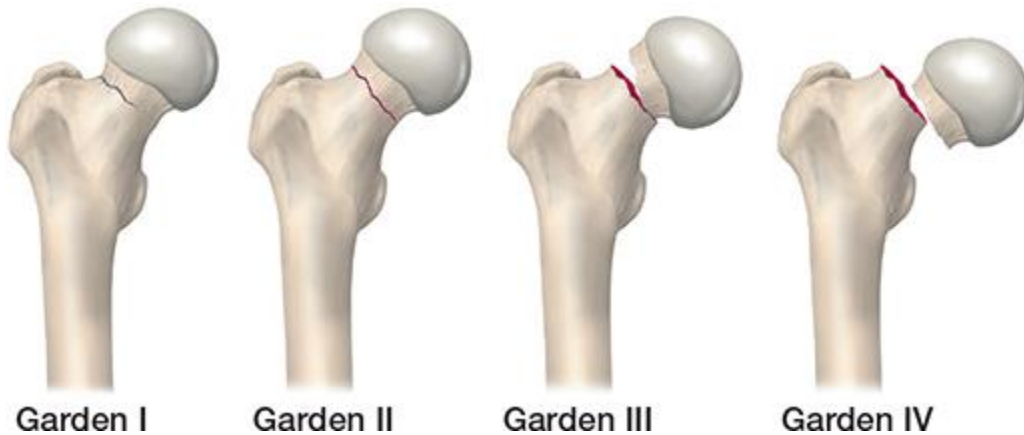
1. Intracapsular Fractures (Inside joint capsule)

- **Subcapital** – just below the head (more proximal)
- **Transcervical** – across the neck (more distal)

2. Extracapsular Fractures (Outside joint capsule)

- **Trochanteric** – through greater/lesser trochanter
- **Subtrochanteric** – below trochanters

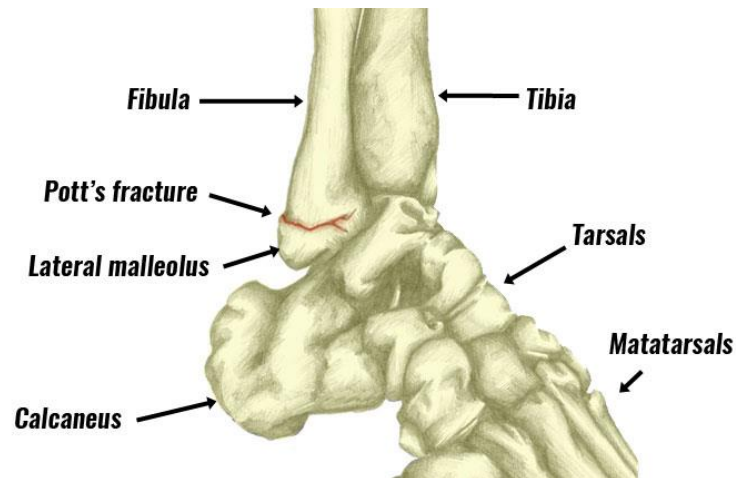
3. Garden Classification (Femoral Neck Fractures)



- **Type I** – Incomplete, impacted; blood supply may be limited → partial necrosis risk
- **Type II** – Complete but undisplaced → risk of necrosis
- **Type III** – Partially displaced
- **Type IV** – Completely displaced

Ankle & Foot Fractures

1. Dupuytren / Pott's Fracture



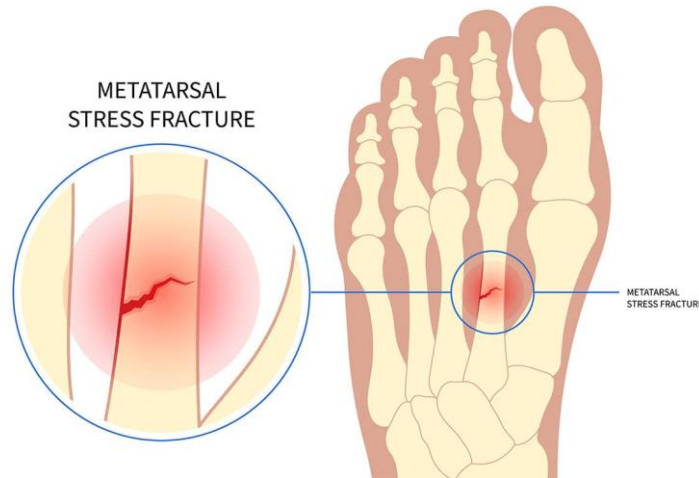
- Fracture of **lower fibula**, often with **medial malleolus fracture** and **deltoid ligament rupture**
- Cause: **Forced eversion of the foot**, affecting **medial collateral ligament**

2. Pilon Fracture



- *Correct definition: Fracture of **distal tibia** involving the ankle joint (Distal metaphysis)*

3. March Fracture / Stress Fracture



- *Common in **metatarsals**, often due to repetitive stress*

Foot Deformities

1. Flat Foot (*Pes Planus*)

- *Arch collapse due to weakened ligaments that normally maintain the medial arch*

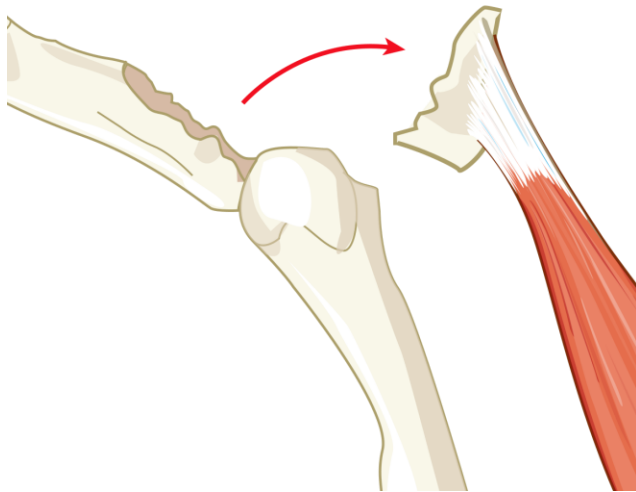
2. Claw Foot (*Pes Cavus*)

- *Excessively high medial arch*

Latin Vocabulary:

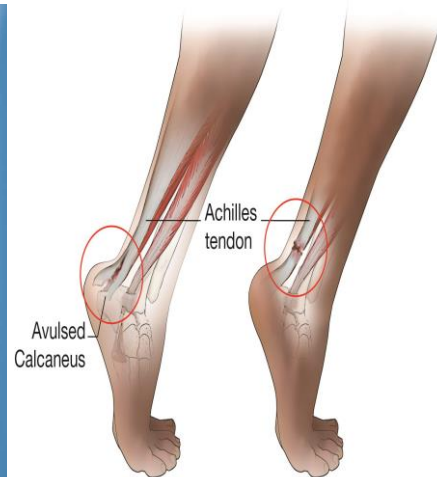
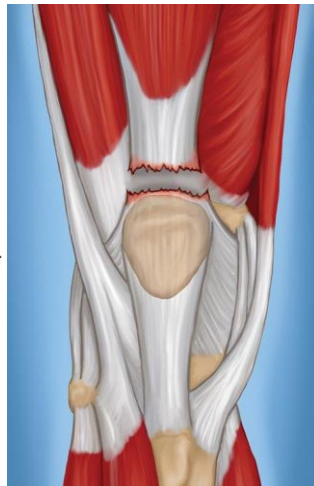
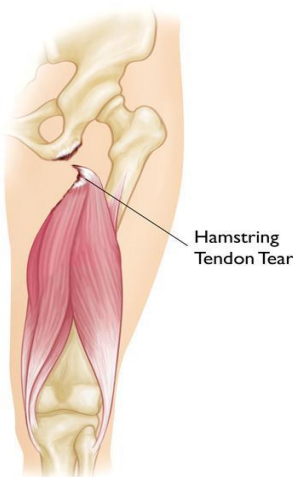
- ***Pes*** = foot
- ***Planus*** = flat
- ***Cavus*** = hollow, cave-like

Avulsion Injuries



This is Avulsion

- **Excessive extension of thigh** → *Hamstrings avulsion*
- **Excessive knee flexion** → *Quadriceps femoris avulsion*
- **Excessive dorsiflexion at ankle** → *Achilles (calcaneal) tendon avulsion*



Superficial and Deep Fascia of the Lower Limb

1. Superficial Fascia

- *Lies directly beneath the skin.*
- *Contains fat, cutaneous nerves, superficial veins, and lymphatics.*

2. Deep Fascia

- *Strong, dense connective tissue that **invests the limb like an elastic stocking.***
- *Functions:*
 - *Supports muscles*
 - *Provides compartmentalization*
 - *Forms a rigid boundary limiting expansion*

Examples:

- ***Fascia lata** → deep fascia of the thigh*
- ***Crural fascia** → deep fascia of the leg*

3. Intermuscular Septa

- *Extensions of deep fascia that **divide muscles into compartments.***
- *Example:*
 - *Thigh → 3 compartments (Extensors, Adductors, Flexors)*
 - *Leg → 3–4 compartments (Extensors, Flexors, and sometimes lateral compartment)*
- *Function:*
 - *Keeps muscles grouped and organized*
 - *Forces muscles to push against fascia during contraction*

Compartment Syndrome

Mechanism:

- Occurs when pressure **inside a fascial compartment increases**, compressing muscles, arteries, veins, and nerves.
- Causes:
 - Trauma, swelling, bleeding, tight fascia
 - External pressure (e.g., tight casts, prolonged sitting with wallets in pockets)

Consequences:

- Compression of **neurovascular structures** → ischemia, nerve injury
- **Deep veins** are particularly affected because their walls are thin and collapsible
- Impaired blood return can reduce the **musculovenous pump** function

Key Points:

- Leg fascia is very tight → compartment syndrome more common here
- Infection or edema increases intracompartmental pressure → harm to muscles & neurovascular bundles
- **Deep fascia defects** like **saphenous hiatus** allow communication between superficial and deep veins

The structure is the iliotibial tract (IT band)

*It is not a muscle. It is a **thickened lateral part of the fascia lata (deep fascia of the thigh).***

Definition (exam-safe):

*The iliotibial tract is a **thickened band of fascia lata on the lateral side of the thigh**, extending from the **iliac tubercle of the iliac crest** down to the **tibial tubercle (specifically Gerdy's tubercle) on the lateral condyle of tibia.***

Now, to your key question:



Which muscles attach to the iliotibial tract?

Two muscles insert into it:

- 1. **Tensor fasciae latae***
- 2. **Gluteus maximus***

*So yes—**two muscles**, exactly as you identified in the picture.*

What happens when these muscles contract?

When **tensor fasciae latae** and **gluteus maximus** contract, they **tense the iliotibial tract**. That tension:

- **Stabilizes the knee joint**, especially in extension
- **Prevents lateral displacement** of the knee
- Helps maintain the **upright standing posture**, especially during single-leg stance (walking, standing on one leg)

This is why the IT tract is often described as a **passive stabilizer of the knee**.

Why is the IT tract important functionally?

Because it:

- Reinforces the **lateral side of the knee**
- Protects the **lateral compartment of the knee joint**
- Acts like a tension cable running down the thigh

Now let's untangle the **fascial compartments** part.

Deep fascia and compartments of the thigh

The **fascia lata** sends **intermuscular septa** to the femur, dividing the thigh into **three compartments**:

1. Anterior compartment

- Muscles: **Quadriceps femoris**, sartorius
- Function: **Extension of knee**, flexion of hip
- Nerve: **Femoral nerve**

2. Medial compartment

- Muscles: **Adductors of hip**
- Function: **Adduction of hip**
- Nerve: **Obturator nerve**

3. **Posterior compartment**

- Muscles: *Hamstrings*
- Function: **Extension of hip**, *flexion of knee*
- Nerve: *Sciatic nerve*

The **iliotibial tract** lies on the **lateral side** and is part of the **deep fascia**, not a compartment itself.

Final correction to keep clear

- *Iliotibial tract $\neq$ muscle*
- It is a **thickened fascia lata**
- **Two muscles attach:** *gluteus maximus + tensor fasciae latae*
- Their contraction **tenses the IT tract**
- This **stabilizes the knee in standing and walking**

THE END OF LECTURE 2

Ana Mohammed